Bank Telemarketing

CIND119 - Group 2

Carman Fan, carman.fan@torontomu.ca
Carlos Elizondo, celizondo@torontomu.ca
Thanos Paschalis, thanasios.paschalis@torontomu.ca

Toronto Metropolitan University

Introduction

- Aim of the Project:
  - A Portuguese bank seeks to improve its telemarketing strategy to identify customers likely to subscribe to a long-term deposit account
- Why Machine Learning?
  - Phone-based campaigns require predicting subscription likelihood in advance to reduce cost and improve conversion rates
- **Goal:** apply data analytics and machine learning classification to support data-driven targeting decisions

Toronto Metropolitan University

Methodology

- **Tools:**
  - Python (pandas, scikit-learn, matplotlib, seaborn)
- **Machine Learning Methods:**
  - Decision Tree (DT) Classification
  - Naïve Bayes (NB) Classification
- **Strategy for Dataset Split:**
  - Train-Test Split (80/20, stratified sampling)
- **Evaluation Metrics:**
  - Confusion Matrix (True Positive Rate, and False Positive Rate)
  - Accuracy, Recall, Precision, F1-Score
  - Classification Report

Toronto Metropolitan University

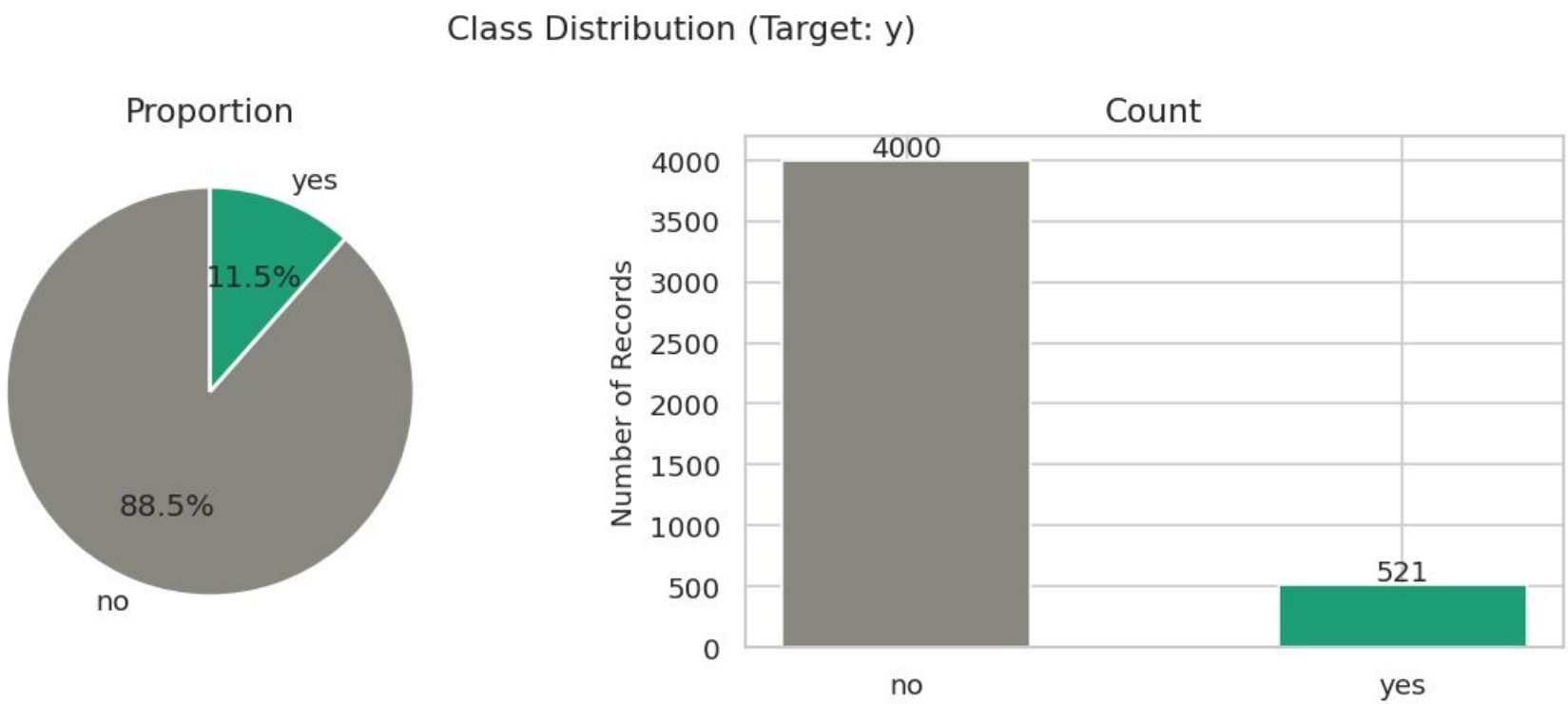
Dataset Summary

- Bank Marketing Dataset
  - 4,521 records (rows)
  - 17 attributes (columns)
  - Target variable: y — whether the customer subscribed to a term deposit (yes/no)

Attribute	Type	SubType	Note	Description
Demographics				
Age	Qualitative	Ordinal	numeric	Age of the customer
Job	Qualitative	Nominal	qualitative	Type of job
Marital	Qualitative	Nominal	qualitative	Marital status
Education	Qualitative	Ordinal	qualitative	Education of the customer
Customer Profile				
Default	Qualitative	Nominal	qualitative	Shows whether the customer has credit in default or not
Balance	Quantitative	Continuous	numeric	Average yearly balance in Euros
Housing	Qualitative	Nominal	qualitative	Shows whether the customer has a housing loan or not
Loan	Qualitative	Nominal	qualitative/categorical	Shows whether the customer has a personal loan or not
Campaign Behaviour				
Contact	Qualitative	Nominal	qualitative	Shows how the last contact for marketing campaign has been made
Day	Qualitative	Ordinal	numeric	Shows on which day of the month, last time a customer was contacted
Month	Qualitative	Ordinal	qualitative	Shows on which month of the year, last time a customer was contacted
Duration	Quantitative	Discrete	numeric	Shows the last contact duration in seconds
Campaign	Quantitative	Discrete	numeric	Number of contacts performed during the marketing campaign and for this customer
Pdays	Quantitative	Discrete	numeric	Number of days that passed by after the client was last contacted from a previous campaign (-1 means client was not previously contacted)
Previous	Quantitative	Discrete	numeric	Number of contacts performed before this campaign and for this client
Poutcome	Qualitative	Nominal	qualitative	Outcome of the previous marketing campaign
Y	Qualitative	Nominal	qualitative	Class attribute showing whether the client has subscribed a term deposit or not (binary: "yes","no"). People with subscription to term deposit are represented by yes and people with no subscription to the term deposit will be reported by no.

Toronto Metropolitan University

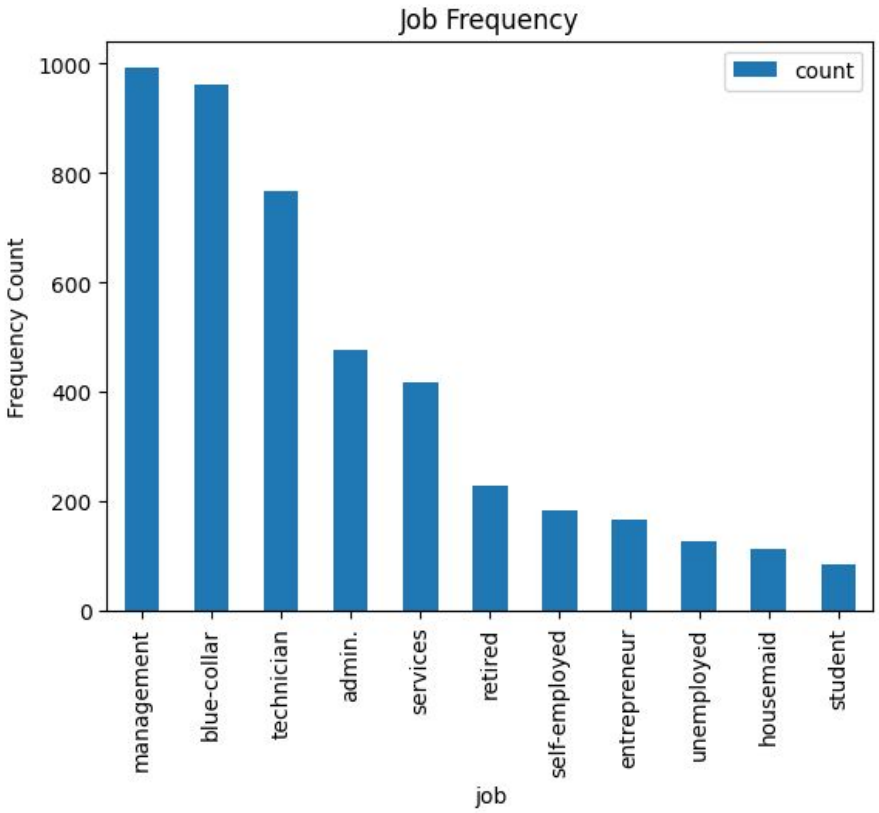
Data Preparation: Attributes & Class Imbalance



Toronto Metropolitan University

Data Preparation: Unknowns

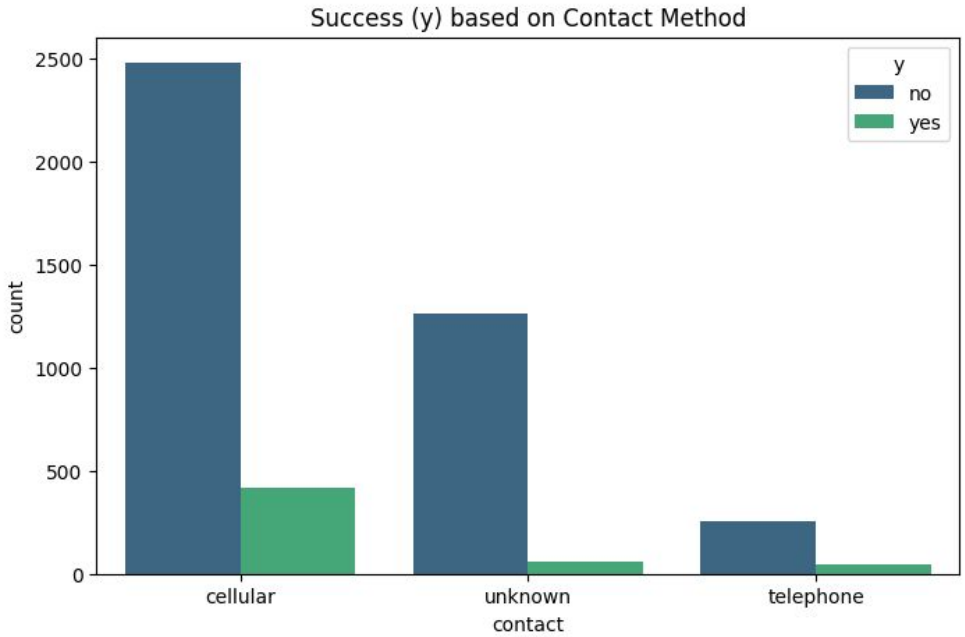
- Columns with “unknown” values:
  - Job: 38 unknowns
  - Education: 187 Unknowns
  - Contact: 1324 Unknowns
  - Poutcome: 3705 Unknowns

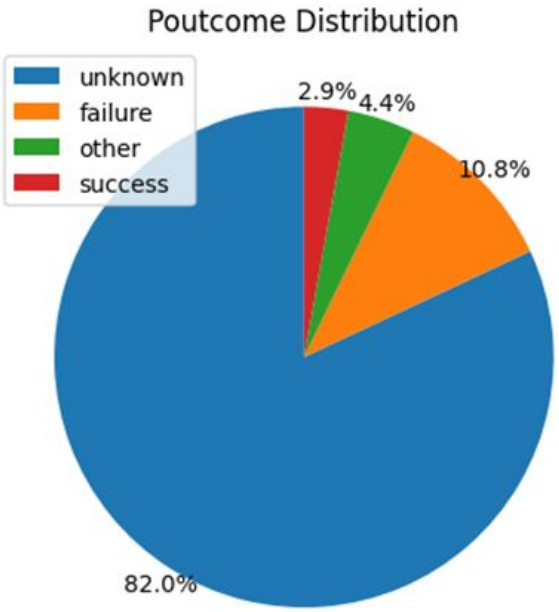


Toronto Metropolitan University

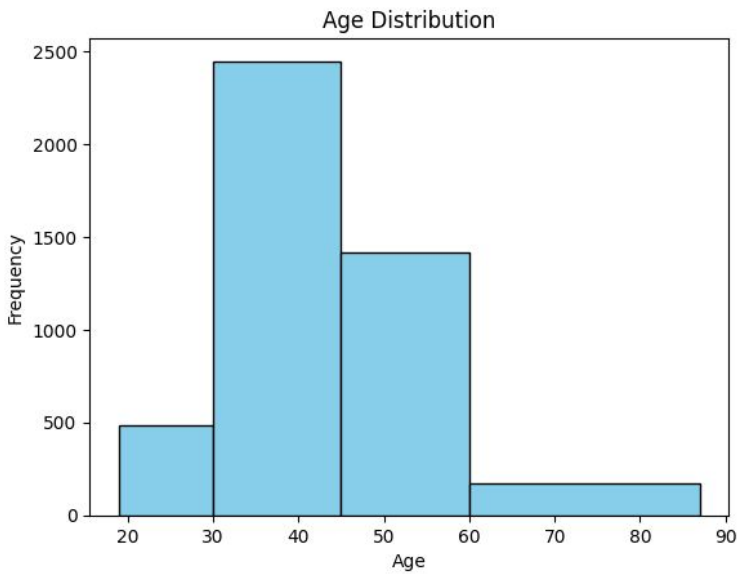
Data Preparation: Unknowns (cont.)

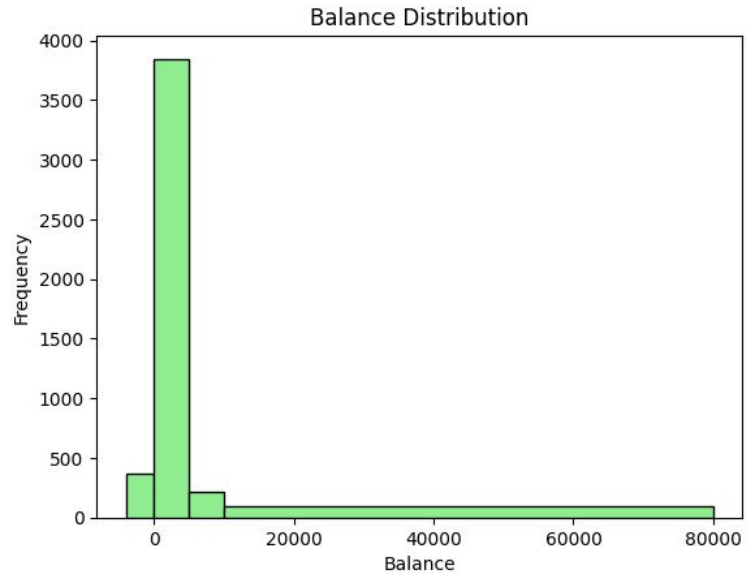
- Columns with “unknown” values:
  - Job: 38 unknowns
  - Education: 187 Unknowns
  - Contact: 1324 Unknowns
  - Poutcome: 3705 Unknowns

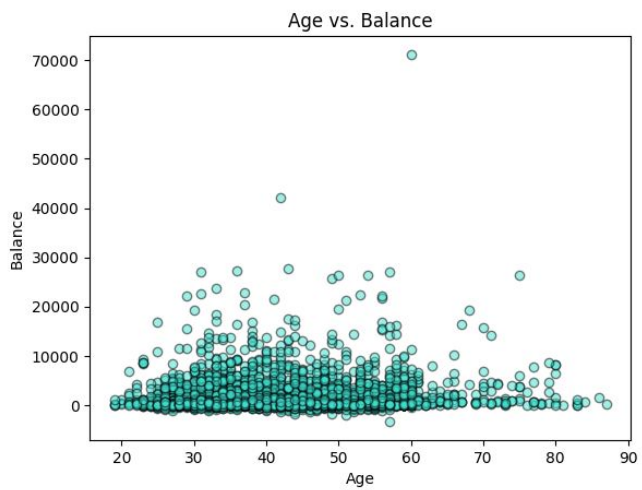




Data Preparation: Attributes Inspection







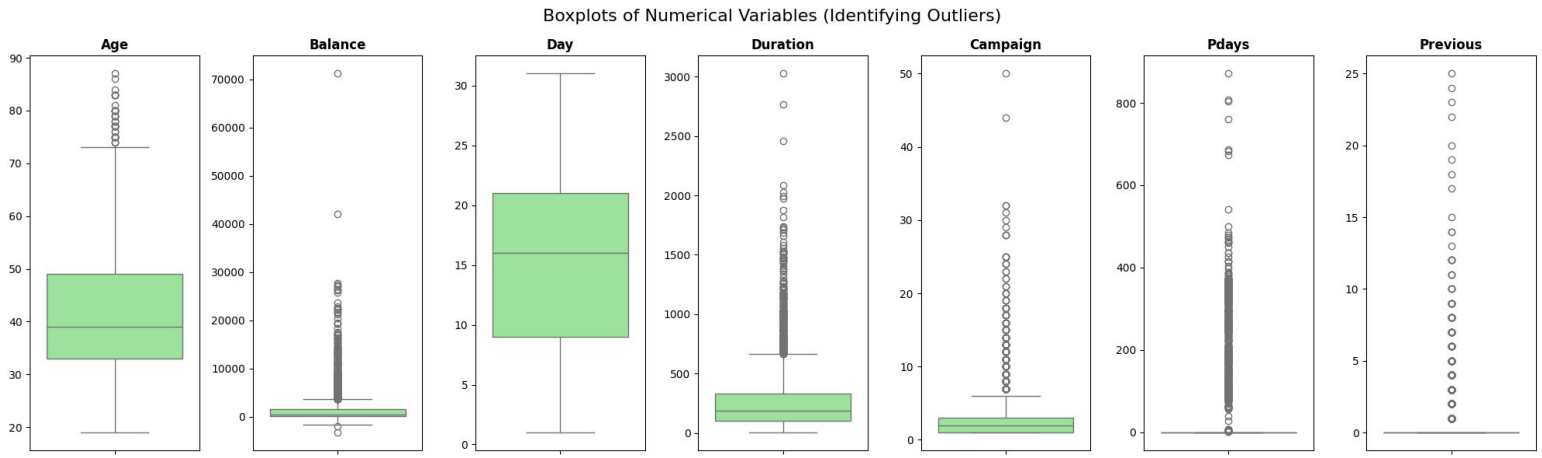
Toronto Metropolitan University

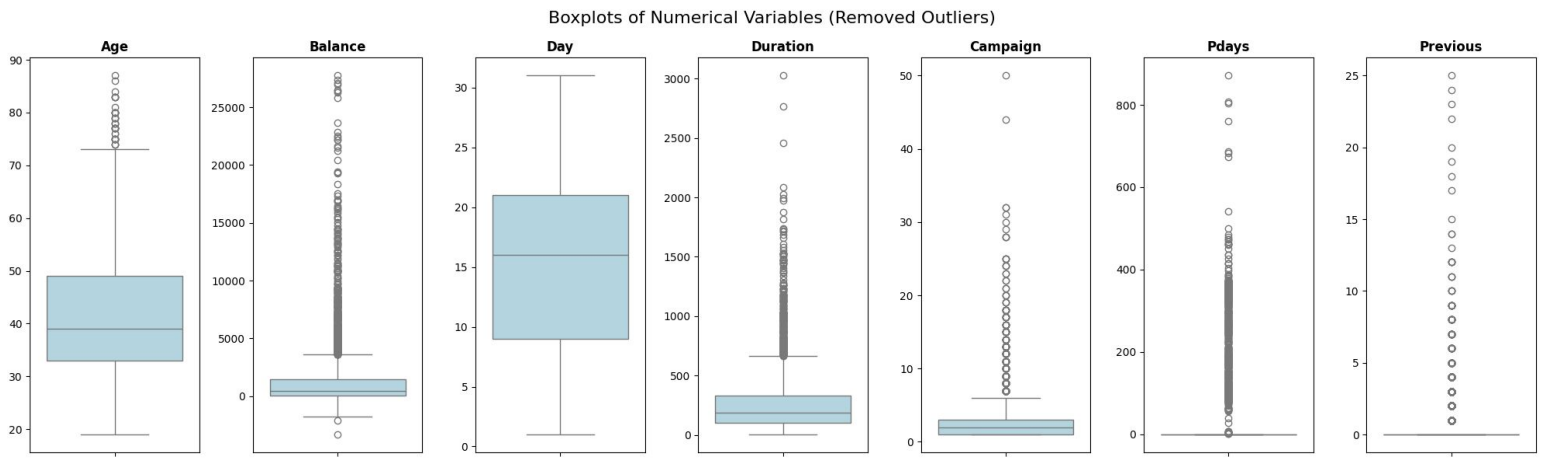
Data Preparation: Measures of Central Tendency

4521 Entries	Age	Balance (€)	Day	Duration	Campaign	pdays	previous
Mean	41.17	1422.66	15.92	263.96	2.79	39.77	0.54
STD	10.58	3009.64	8.25	259.86	3.11	100.12	1.69
MIN	19	-3313.00	1	4	1	-1	0
25%	33	69.00	9	104	1	-1	0
Median	39	444.00	16	185	2	-1	0
75%	49	1480	21	329	3	-1	0
MAX	87	71188.00	31	3025	50	871	25

Toronto Metropolitan University

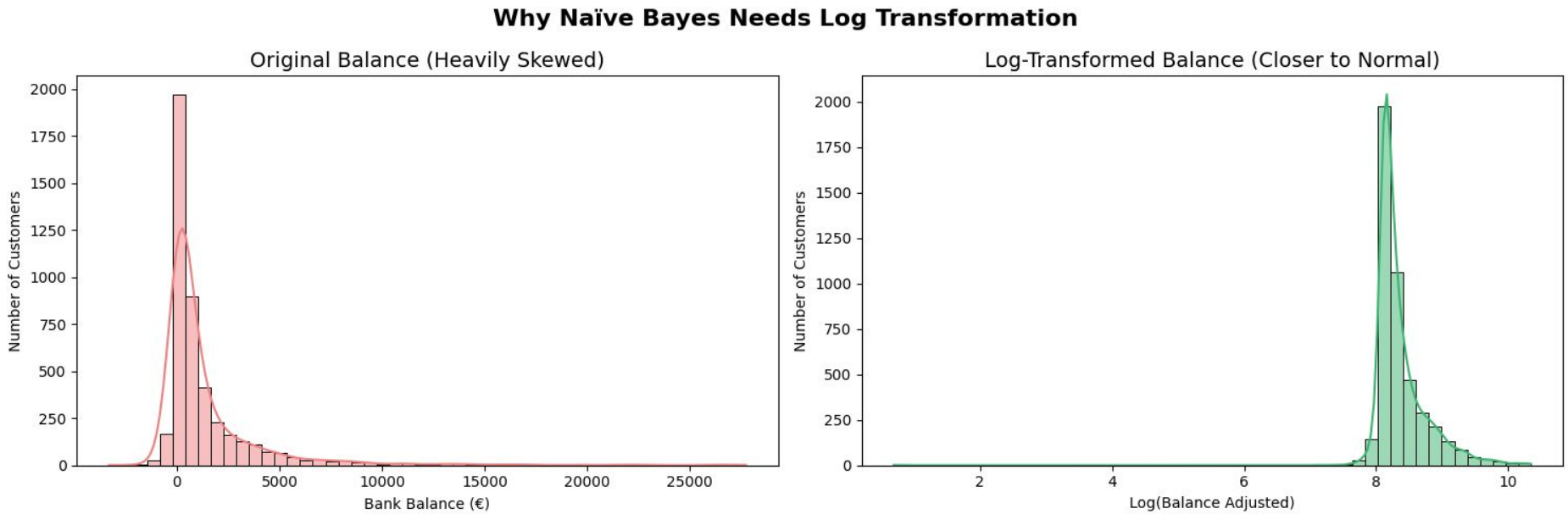
Data Preparation: Outlier Handling





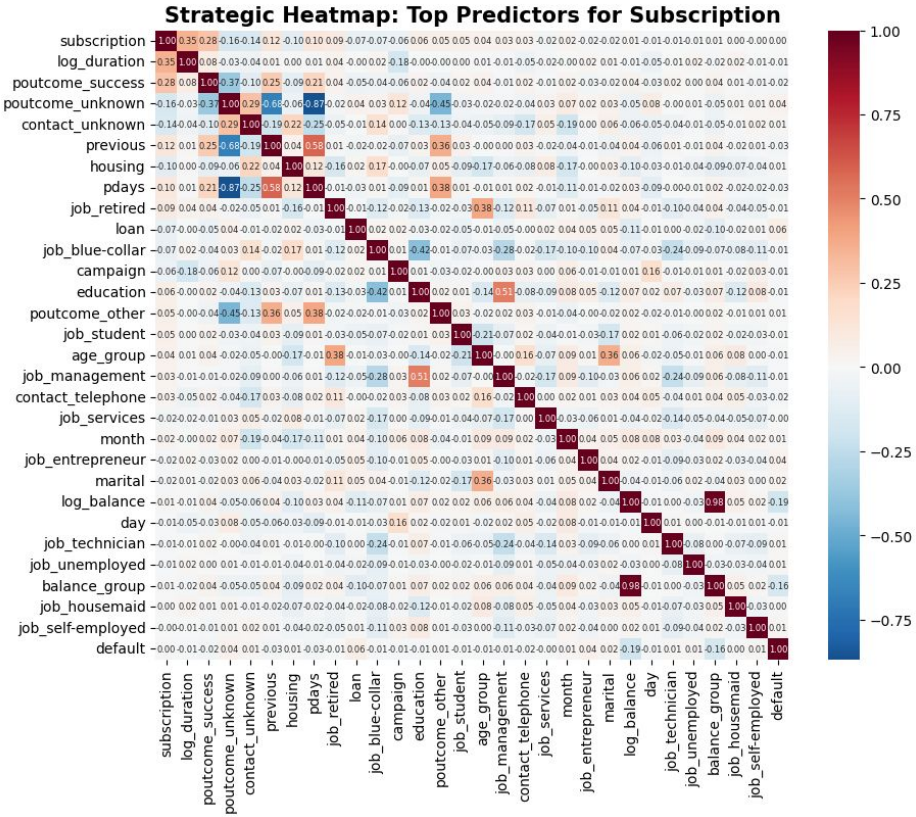
Toronto Metropolitan University

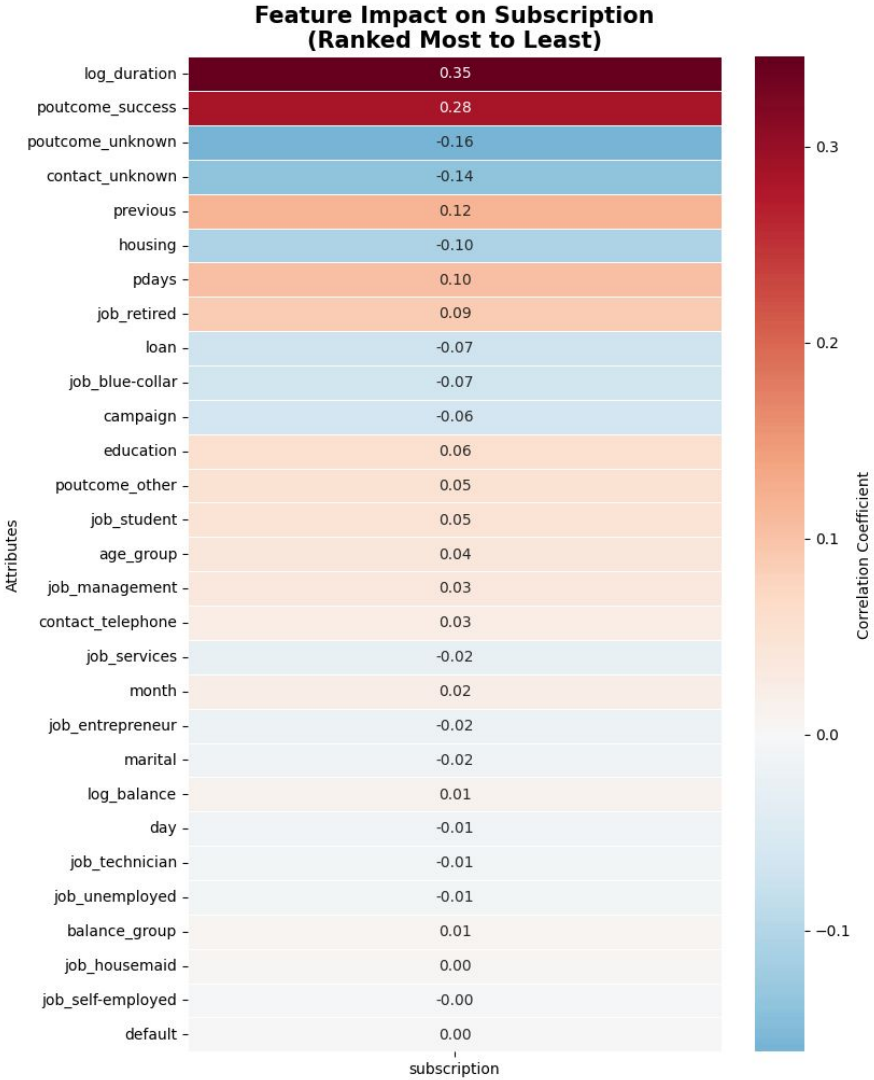
Data Preparation: Log Transformation



Toronto Metropolitan University

Data Preparation: Correlation Analysis





Toronto Metropolitan University

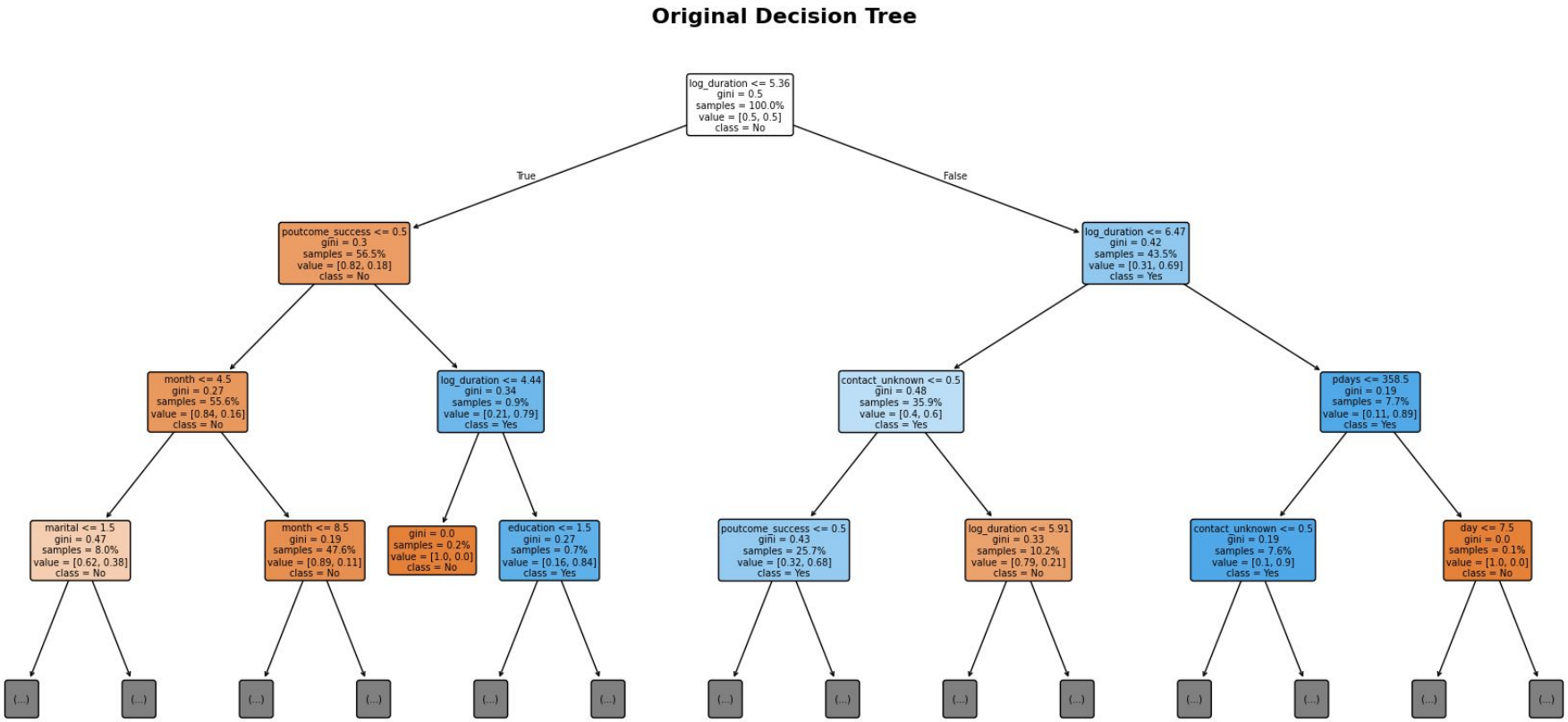
Predictive Modeling

Strategy
80/20 stratified train-test split (random state = 42)

- Training set: 3,615 records
- Test set: 904 records
- Stratified to preserve the 88.5% / 11.5% class ratio in both splits

Why train-test split over cross-validation:
Given the class imbalance, stratified splitting ensures the minority class (subscribed = yes) is proportionally represented in both sets, giving reliable and consistent metric evaluation across all six models.

Predictive Modeling: Baseline Decision Tree



Toronto Metropolitan University

Predictive Modeling: Baseline Naive Bayes

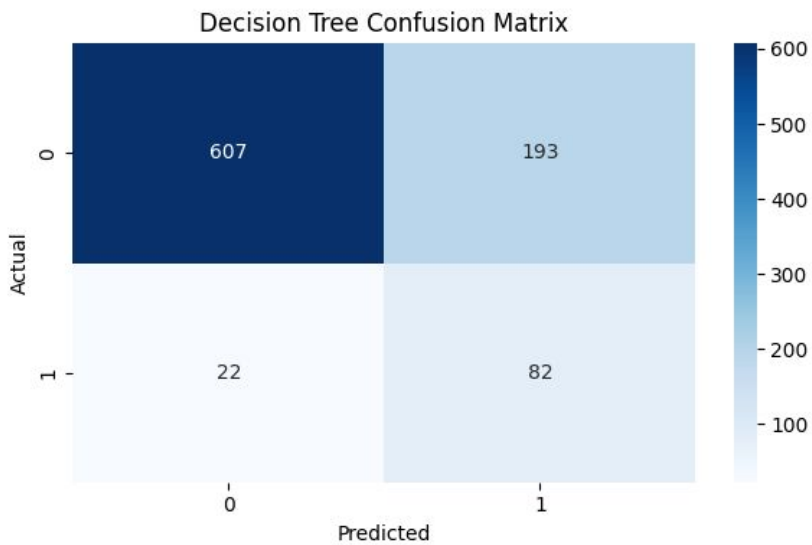
Feature	Permutation Importance	Interpretation
poutcome_success	0.0279	Strongest signal — prior campaign success dramatically increases subscription probability
log_duration	0.0152	Longer calls strongly associated with subscription
job_retired	0.0038	Retired customers more likely to subscribe
housing	0.0034	Having a housing loan is negatively associated with subscription
loan	0.0031	Having a personal loan is negatively associated with subscription
job_student	0.0019	Student customers show above-average subscription rates

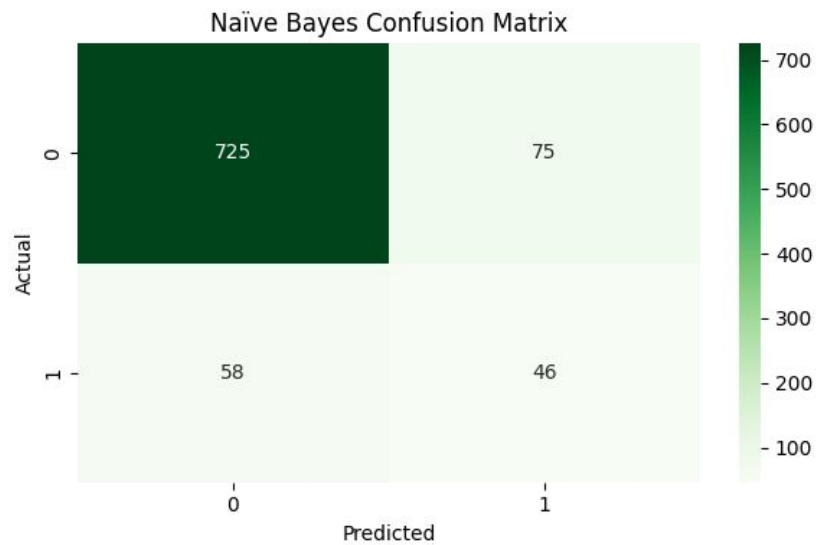
Toronto Metropolitan University

Predictive Modeling:

Baseline Models Evaluation - Preliminary Results

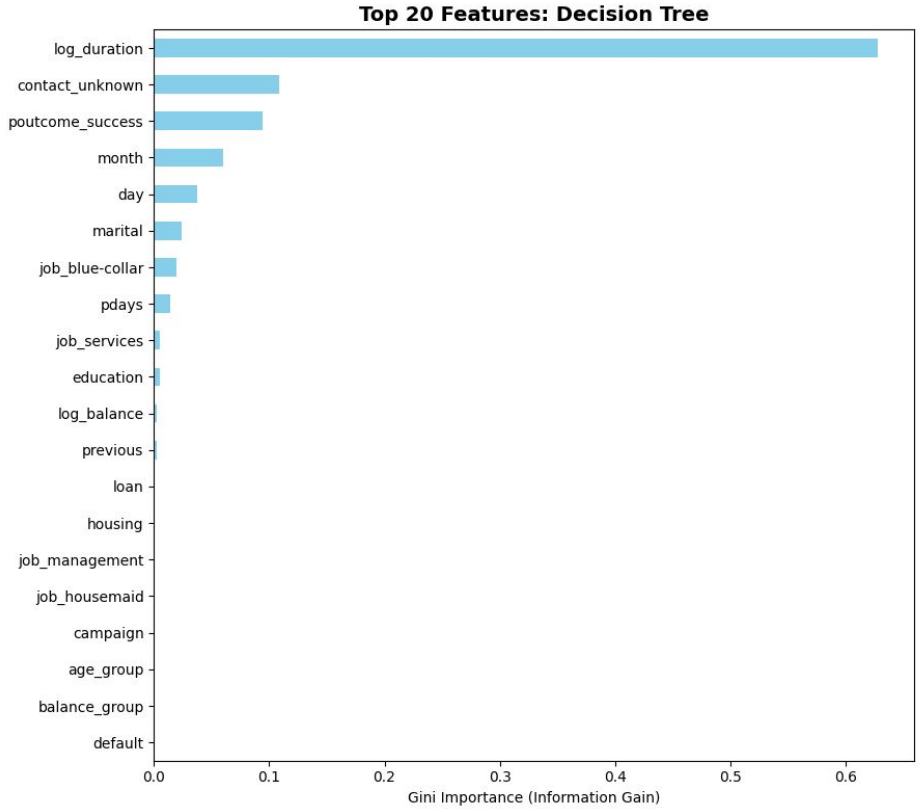
	Accuracy	Precision	Recall	FPR	F-1 Score
Decision Tree Baseline	0.7622	0.2982	0.7885	0.2412	0.4327
Naive Bayes Baseline	0.8529	0.3802	0.4423	0.0938	0.4089

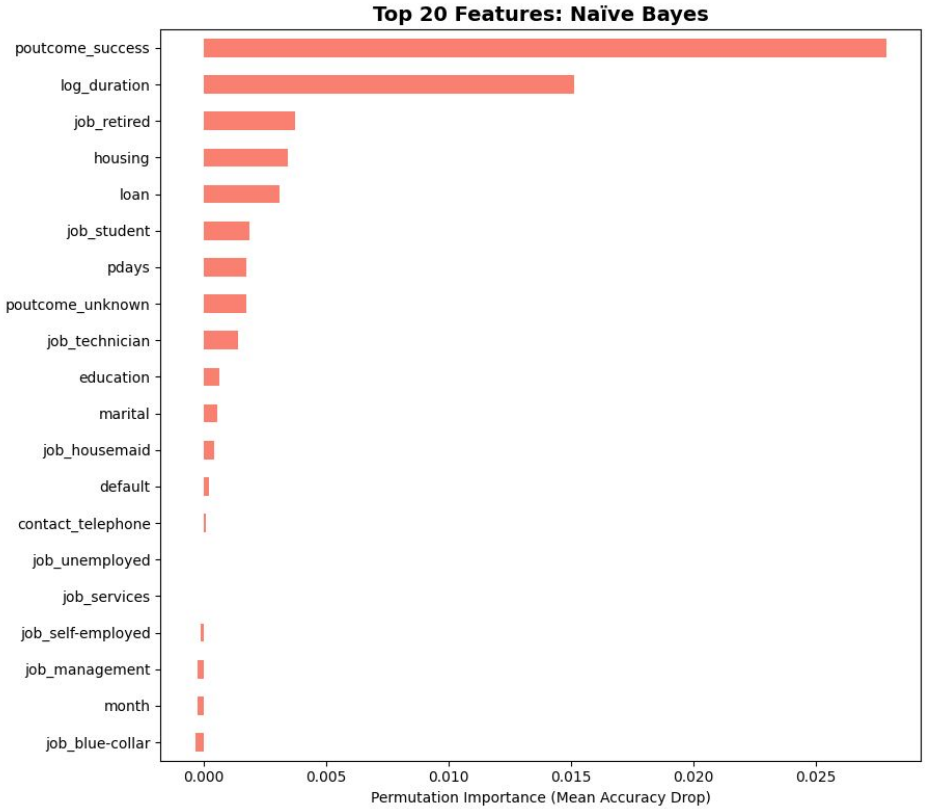




Toronto Metropolitan University

Predictive Modeling: Feature Importance based on Baseline Models



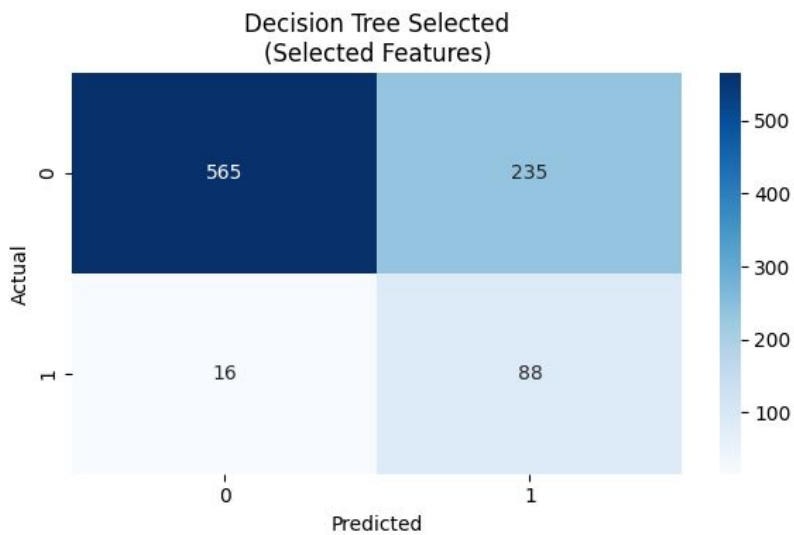


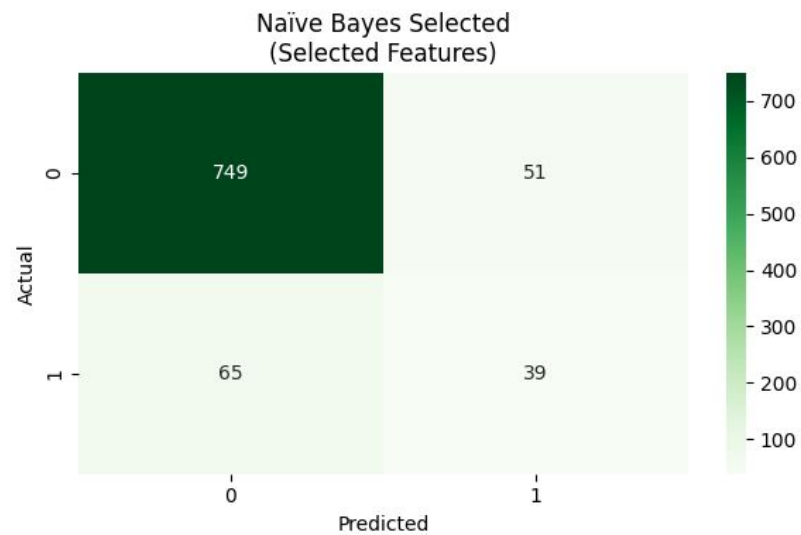
Toronto Metropolitan University

Predictive Modeling:

Selected Features Models Evaluation - Preliminary Results

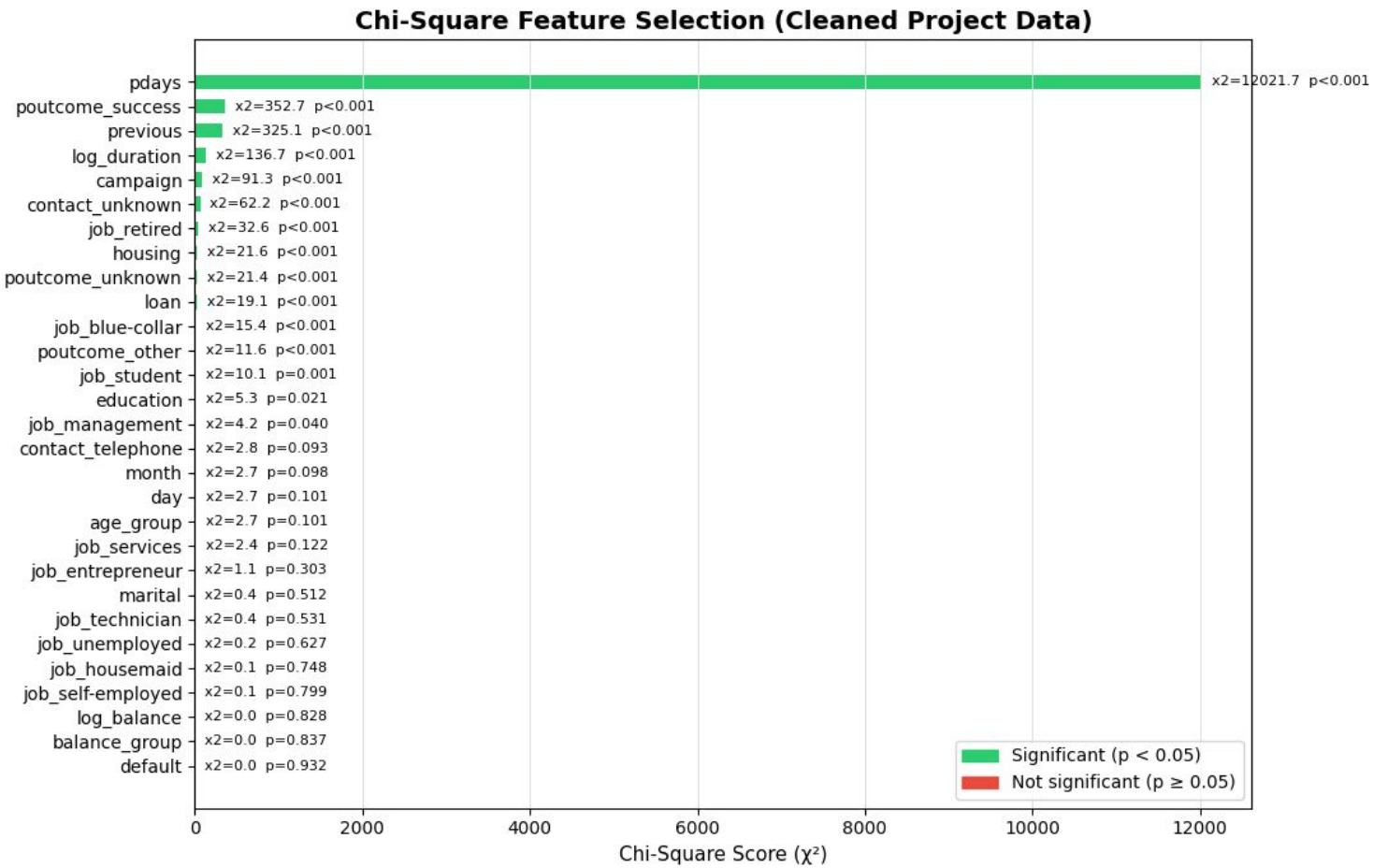
	Accuracy	Precision	Recall	FPR	F-1 Score
Decision Tree Selected Features	0.7223	0.2724	0.8462	0.2938	0.4122
Naive Bayes Selected Features	0.8717	0.4333	0.3750	0.0638	0.4021





Toronto Metropolitan University

Bonus: Chi-Square Analysis



Toronto Metropolitan University

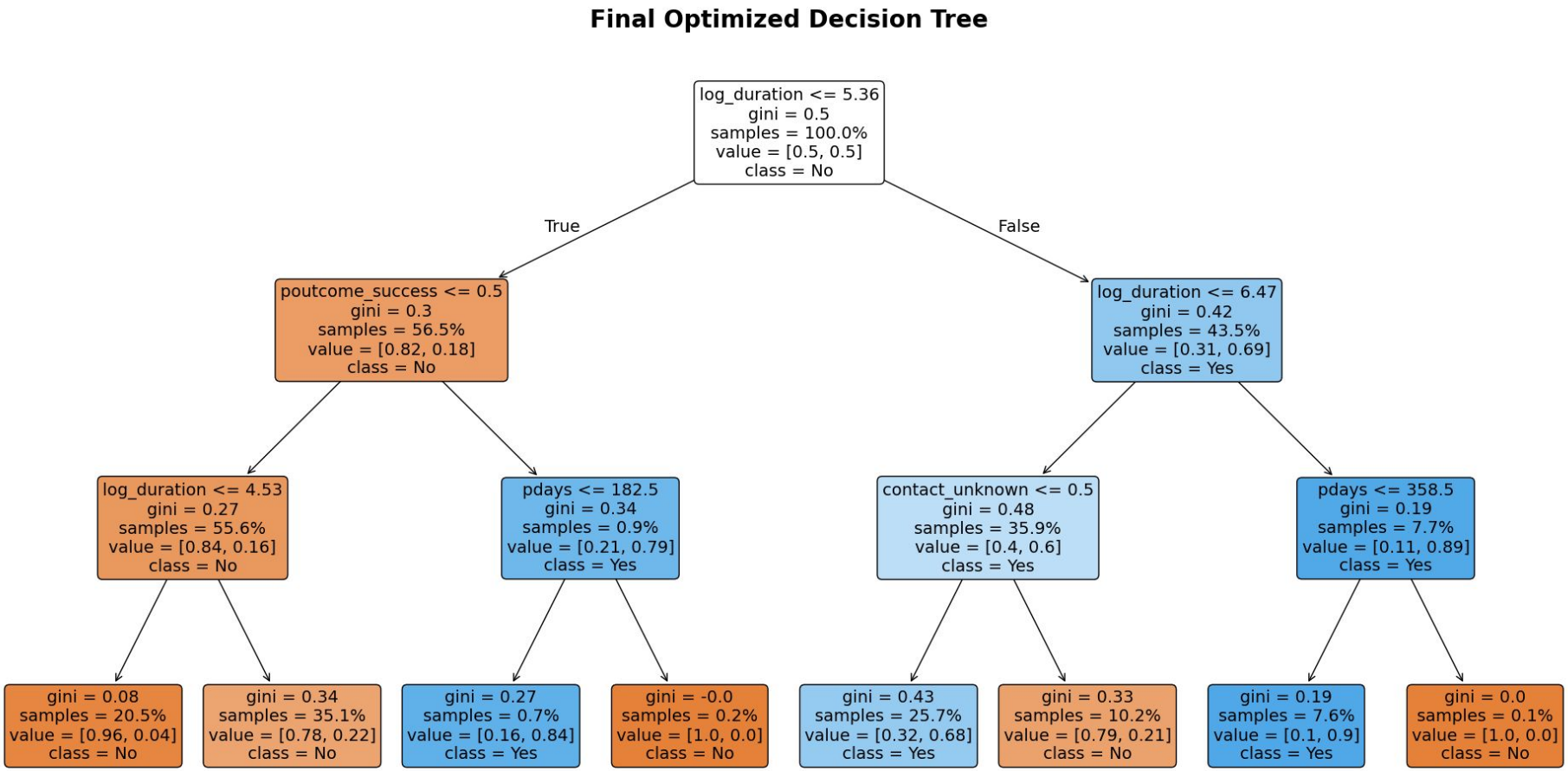
Bonus: Feature Selection via Consensus
Combining Baseline Models with Correlation, Chi-Square

Total Unique Features Selected: 13

	Total_Votes	Golden_Ticket	Correlation	DecisionTree	NaiveBayes	Chi2
poutcome_success	4	⭐	1	1	1	1
log_duration	4	⭐	1	1	1	1
pdays	4	⭐	1	1	1	1
previous	3	⭐	1	1	0	1
poutcome_unknown	3	⭐	1	0	1	1
loan	3		1	0	1	1
job_retired	3	⭐	1	0	1	1
job_blue-collar	3		1	1	0	1
education	3		1	1	1	0
contact_unknown	3	⭐	1	1	0	1
housing	3		1	0	1	1
campaign	2		1	0	0	1
poutcome_other	2		1	0	0	1

Toronto Metropolitan University

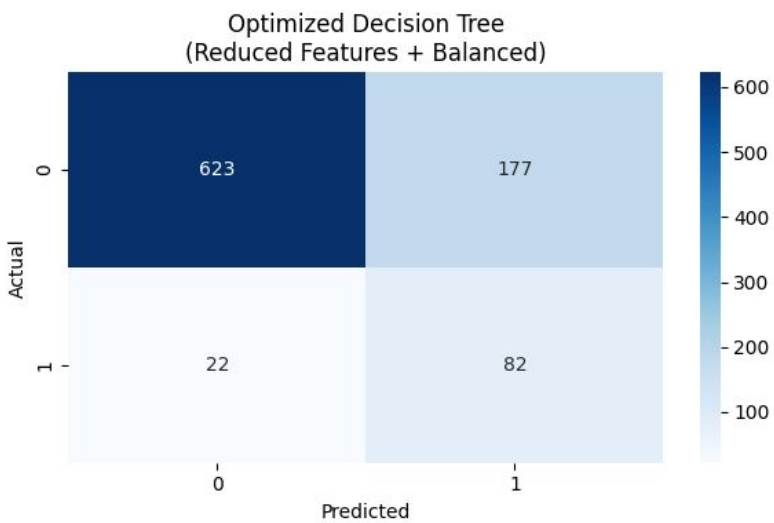
Bonus: Decision Tree w/ Consensus Features

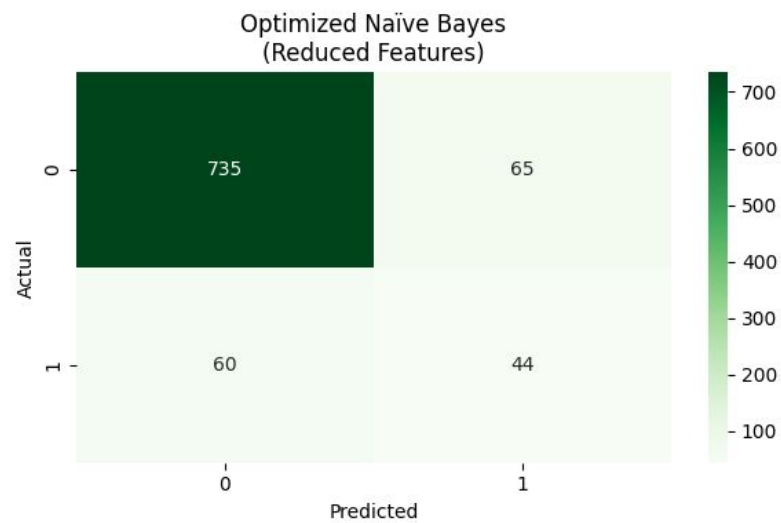


Toronto Metropolitan University

Bonus: Optimized Model Performance w/ Consensus Features

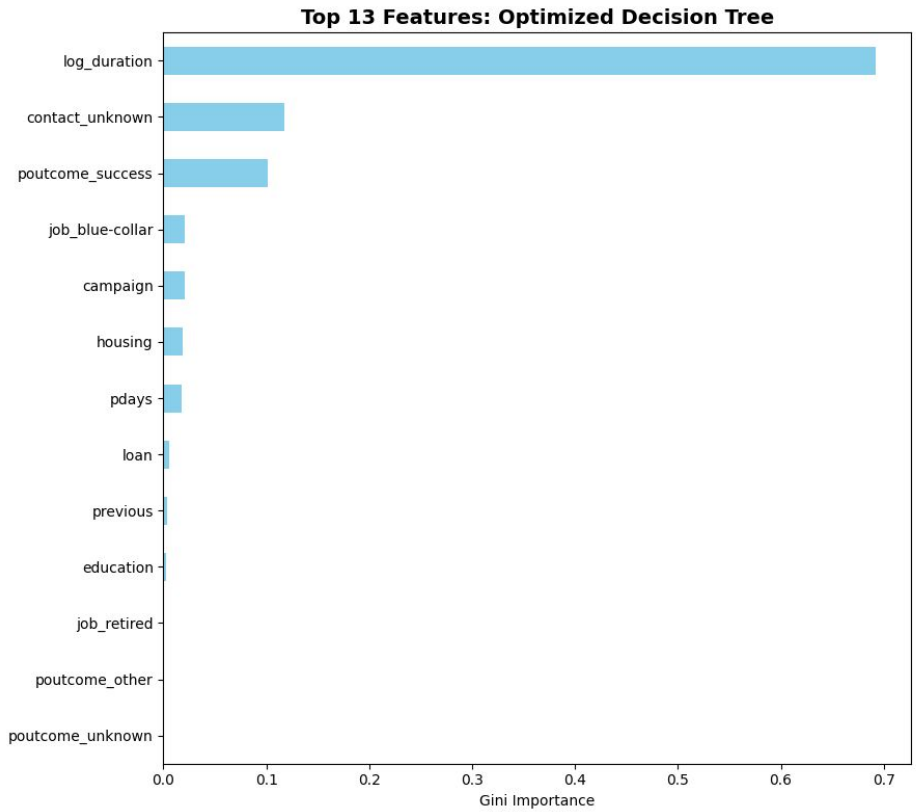
	Accuracy	Precision	Recall	FPR	F-1 Score
Decision Tree Consensus Features	0.7799	0.3166	0.7885	0.2213	0.4518
Naive Bayes Consensus Features	0.8617	0.4037	0.4231	0.0813	0.4131

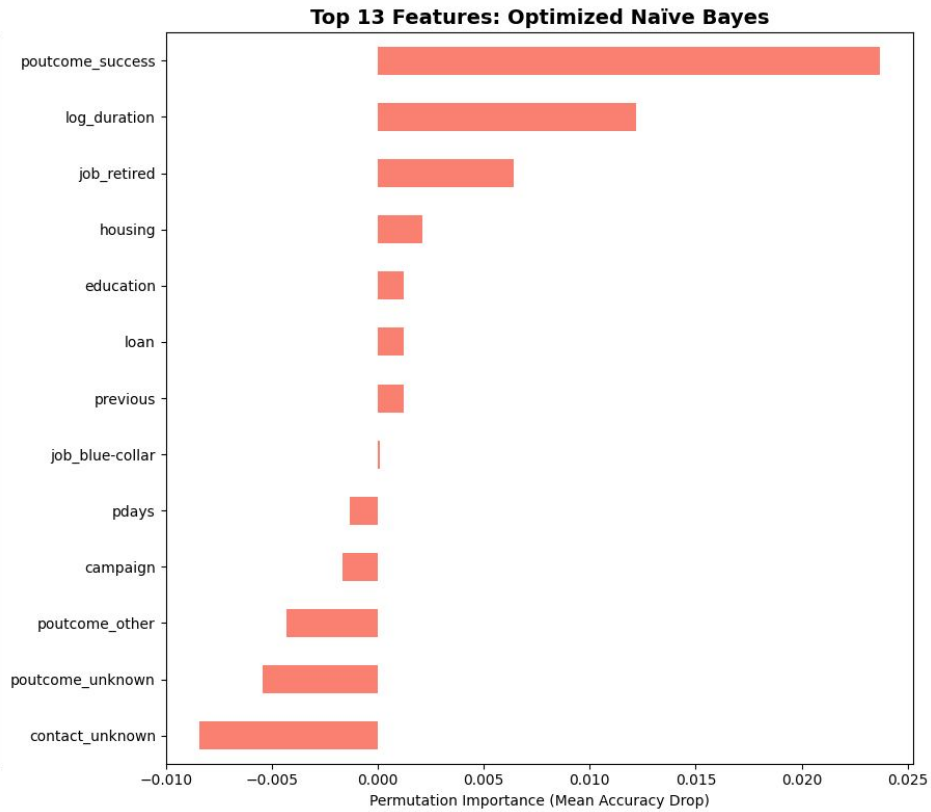




Toronto Metropolitan University

Decision Tree: Selected Features





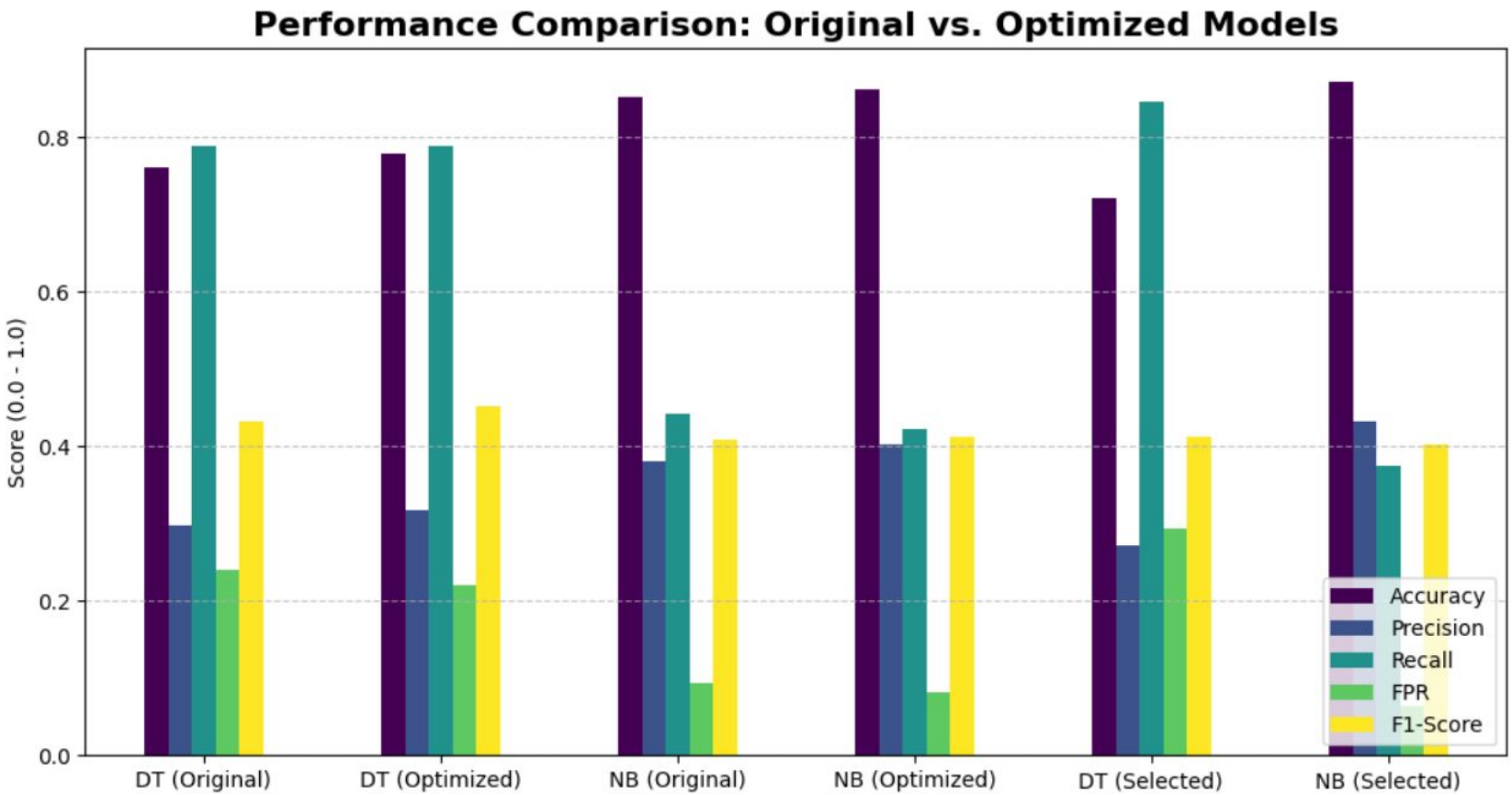
Toronto Metropolitan University

Baseline vs. Selected vs. Consensus Comparison

Comparison	Accuracy	Recall	FPR	F1-Score
DT Original (all 29 features)	76.22%	78.85%	24.12%	0.4327
DT Selected (13 from DT)	72.23%	**84.62%**	29.38%	0.4122
DT Optimized (13 Consensus)	77.99%	78.85%	22.13%	**0.4518**
NB Original (all 29 features)	85.29%	44.23%	9.38%	0.4089
NB Selected (13 from DT)	**87.17%**	37.50%	**6.38%**	0.4021
NB Optimized (13 Consensus)	86.17%	42.31%	8.13%	0.4131

Toronto Metropolitan University

Baseline vs. Selected vs. Consensus Comparison



Toronto Metropolitan University

Major Findings: Data Preparation

- **The dataset is significantly imbalanced** — 88.5% of customers did not subscribe, only 11.5% did. This makes accuracy a misleading metric and elevates TPR and FPR as the decisive evaluation criteria throughout the project.

- **Outlier analysis** revealed that **balance**, **pdays**, **previous**, **duration**, and **campaign** all contain substantial outliers — these reflect real customer behaviour (e.g., high-balance clients, heavily contacted customers) and were deliberately retained as meaningful signals rather than noise.

- **Chi-square analysis** was applied to all 29 prepared features (including log_duration, one-hot encoded job, contact, and poutcome dummies) and combined with three other selection methods into a 4-method Voting Consensus. The top features confirmed by chi-square include log_duration, poutcome_success, pdays, and contact_unknown — consistent with all other selection methods.

- **log_duration (log-transformed call duration)** was the single strongest predictor across every analysis method — Chi-Square (highest χ^2 score), Pearson correlation ($r = 0.40$ on raw duration), and Decision Tree Gini importance (62.8% of total information gain in the baseline model). Call duration is the most reliable signal in the dataset.

Toronto Metropolitan University

Major Findings: Predictive Modeling

- Six models were trained and evaluated across three strategies: baseline (all 29 features), selected (13 features from DT importance), and optimized (13 Voting Consensus features).

- The **Decision Tree Baseline** achieved a **TPR of 78.85%** — correctly identifying nearly 4 in 5 actual subscribers — but at the cost of a high FPR of 24.12%, meaning a large number of non-subscribers would also be called. Accuracy was 76.22%.

- The **Naïve Bayes Baseline** achieved the **best accuracy at 85.29%** with a significantly lower FPR of 9.38%, but a lower TPR of 44.23% — identifying fewer subscribers overall.

- The **DT Optimized (Consensus 13 features) delivered the best F1-Score (0.4518)**, with TPR of 78.85%, FPR of 22.13%, and accuracy of 77.99% — an improvement over the DT Baseline across all three rubric metrics.

- The **NB Selected (13 features from DT) achieved the highest accuracy overall (87.17%) and the lowest FPR (6.38%)**, making it the most precise model for avoiding wasted calls, though TPR drops to 37.50%.

- The **DT Selected (13 from DT) achieved the highest TPR of all six models (84.62%)** — the strongest subscriber detection — though this comes with the highest FPR (29.38%).

Toronto Metropolitan University

Conclusion: Which ML to use?

The recommended operational model depends on the bank's **campaign objective**:

For maximum subscriber detection (highest TPR):
Deploy DT Selected — TPR 84.62%, Accuracy 72.23%, FPR 29.38%
Best when the cost of missing a subscriber is higher than the cost of a wasted call

For minimum wasted calls (lowest FPR):
Deploy NB Selected — FPR 6.38%, Accuracy 87.17%, TPR 37.50%
Best when agent time is the primary operational cost

For best overall balance (highest F1-Score):
Deploy DT Optimized (Consensus) — F1 0.4518, TPR 78.85%, FPR 22.13%, Accuracy 77.99%
Recommended as the primary model — it improves on the baseline across all three rubric metrics simultaneously

Toronto Metropolitan University

Recommendations

- **Prioritise call duration as the primary campaign signal.**
  - Aim to engage customers longer during calls
  - Calls that reach up to 645 seconds (apprx. 10 mins) are more likely to be successful
- **Target customers with a prior successful campaign outcome first.**
  - Customers with **poutcome = success** converted at 64.3%
- **Address class imbalance in future modelling cycles.**

Toronto Metropolitan University

Bonus 2: K-Means Customer Segmentation

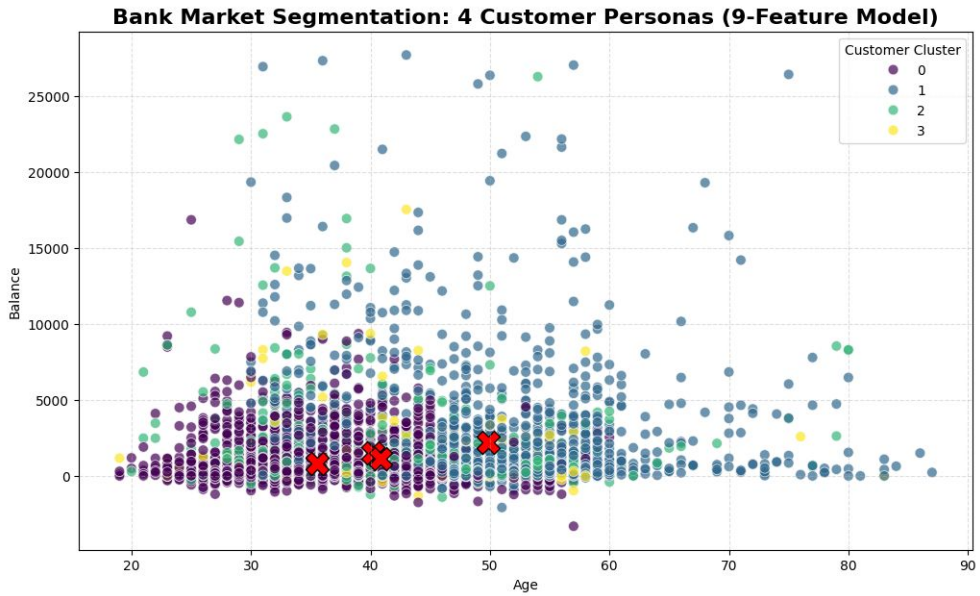
K-Means Clustering (k=4) applied to 9 demographic and behavioural features.

Identified four distinct customer personas with significantly different subscription rates.

Key differentiator for Cluster 2 (Primary Target): last contacted 272 days ago — prior engagement history is the strongest behavioural signal

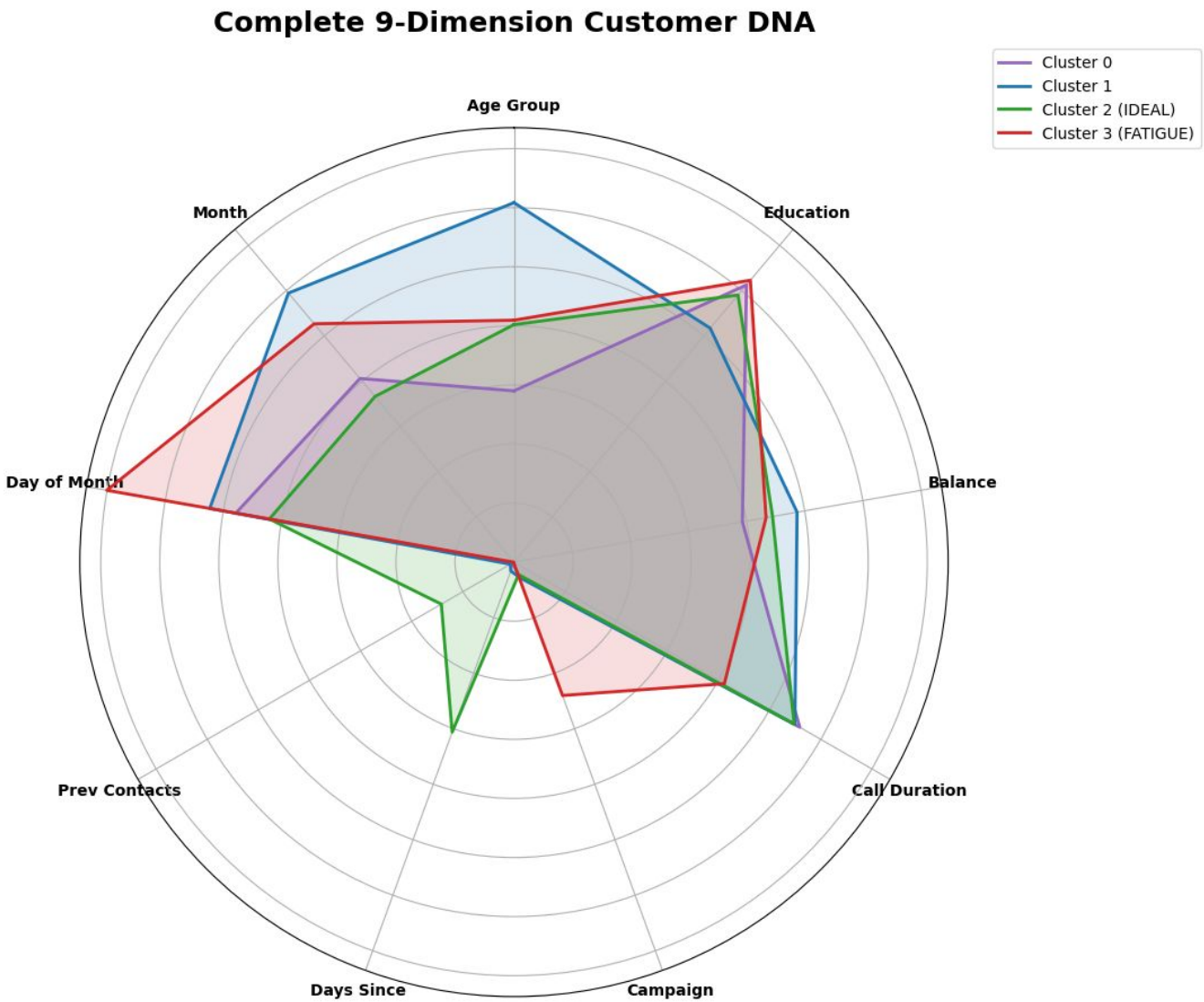
Key warning for Cluster 3: 12.8 average campaign contacts — highest fatigue of any segment

Cluster	Strategy	Avg. Age	Median Balance	Top Job	Success Rate
0	✅ Secondary Target	35 (30-45)	$340	Blue-Collar	10.1%
1	✅ Secondary Target	50 (46-60)	$693	Management	11.8%
2	⭐ Primary Target	38 (30-45)	$510	Blue-Collar	19.6%
3	❌ Cease Contact	39 (30-45)	$340	Management	3.0%



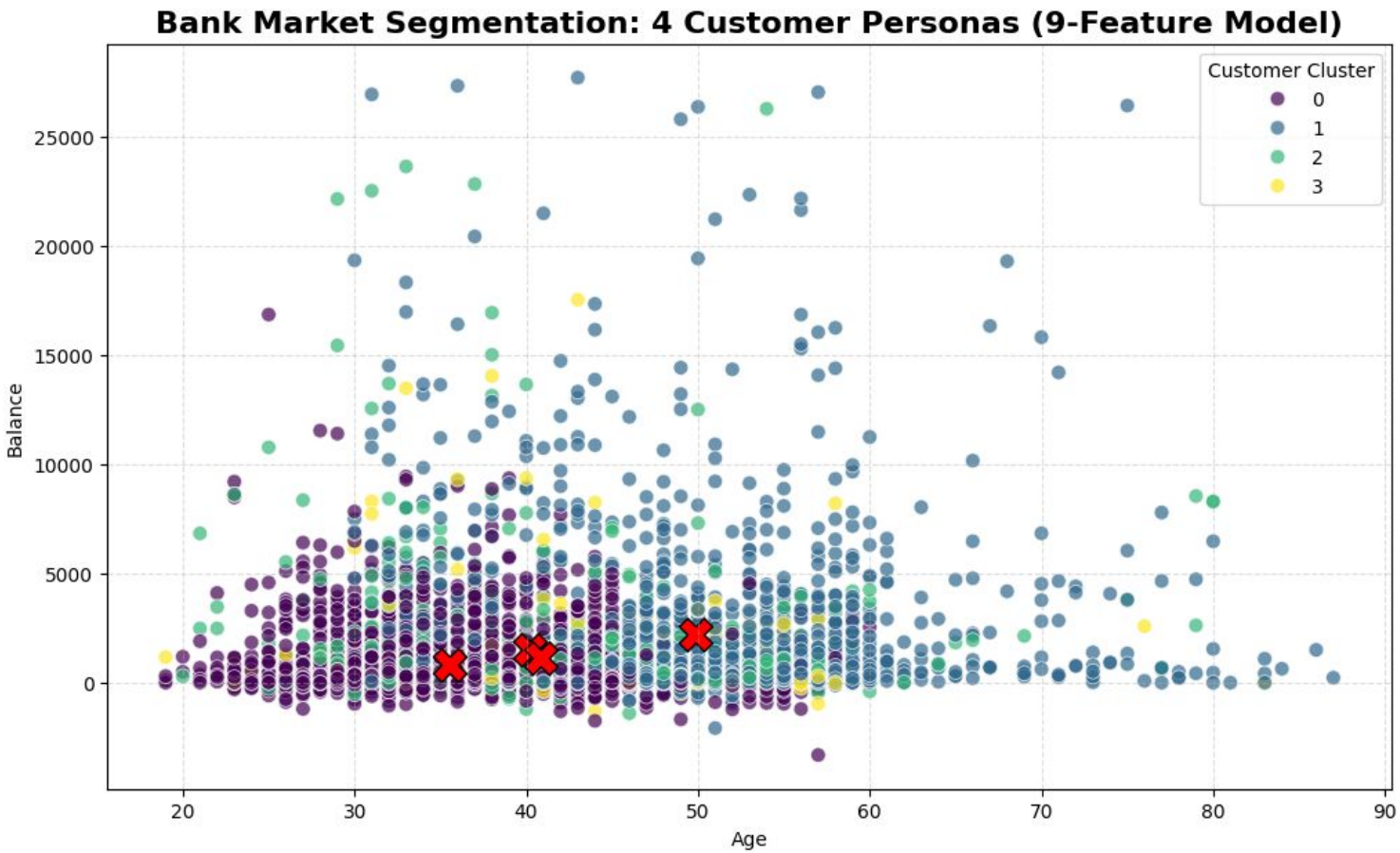
Toronto Metropolitan University

Bonus 2: K-Means Customer Segmentation



Toronto Metropolitan University

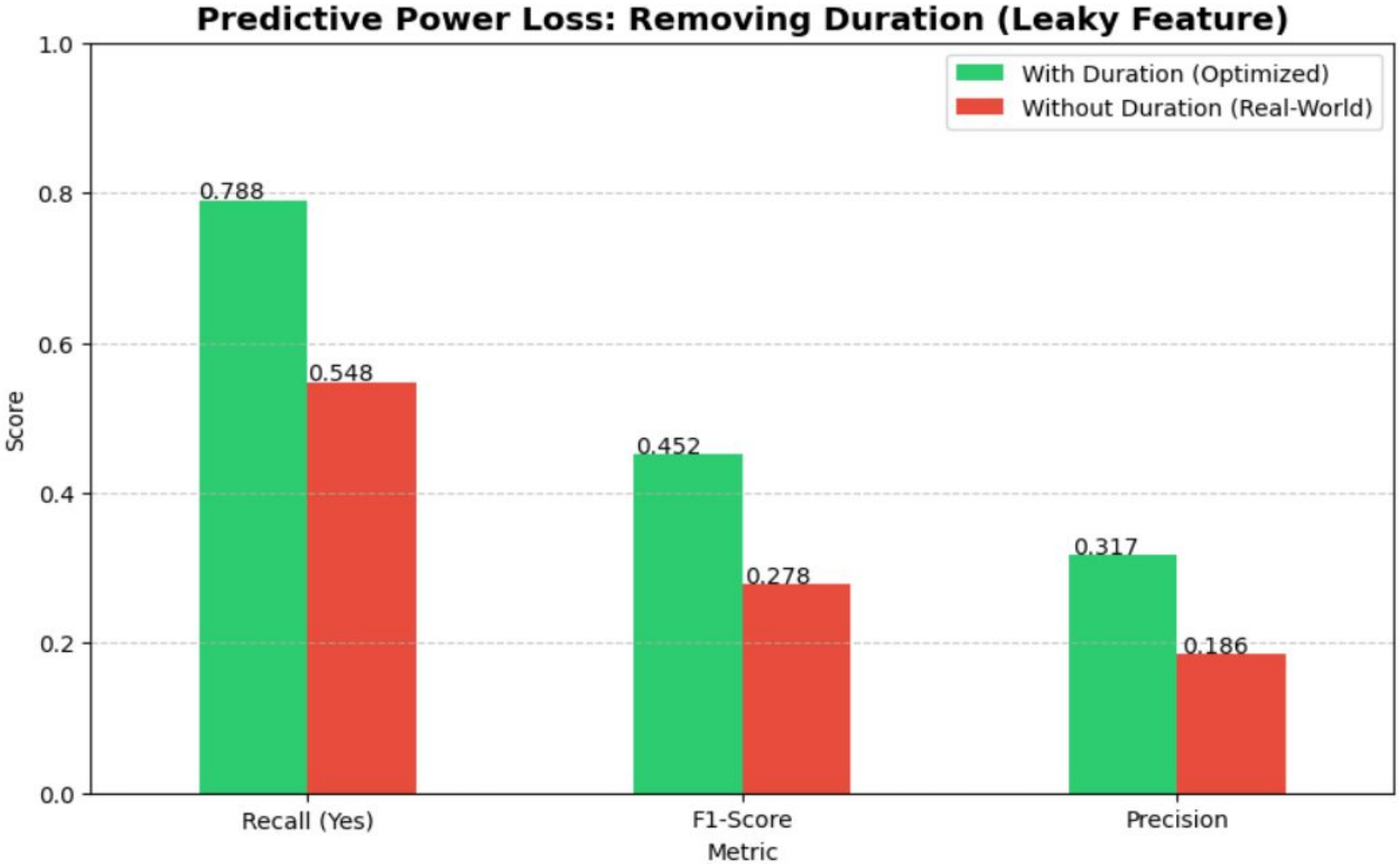
Recommendation (cont'd)



Toronto Metropolitan University

Bonus: Real World Model

```
--- DURATION IMPACT ANALYSIS ---
         Metric  With Duration (Optimized)  Without Duration (Real-World)
0  Recall (Yes)                     0.7885                          0.5481
1      F1-Score                     0.4518                          0.2780
2     Precision                     0.3166                          0.1863
```



Workload Distribution

Member Name	List of Tasks Performed
Carman Fan	Data Preparation. Milestone and Report Authoring.
Carlos Elizondo	Predictive Modelling. Milestone and Report Authoring.
Thanos Paschalis	Statistical Analysis. Milestone and Report Authoring.

Toronto Metropolitan University